Semester: 6 (Even-2026)
Subject: TOC
Assignment Topic: Conversion of Grammar
Targeted CO: CO3
Date: 08/04/2026
Submission Deadline: 22/4/2026

Instructions:
Do as directed.

Prerequisites: Knowledge of FA, NFA, ε-NFA, RG, RE

Step 1. Study Types of Regular Grammar.
- Left Linear Grammar (LLG)
- Right Linear Grammar (RLG)

Step 2. Solve the exercises given below.

Exercise 1. Study equivalence between RLG and ε-NFA and solve the following problems.

1. S -> 01S/1
2. S -> 01S/0/1
3. S -> 11S/01A , A -> 10A/1

Exercise 2. Study equivalence between LLG and ε-NFA and solve the following problems.

1. S -> S10/0/1
2. S -> S11/A10 , A -> A01/1

Exercise 3. Study equivalence between RG (LLG/RLG) and NFA and solve the following problems.

1. S -> 0S/1A , A -> 1B/0A/1 , B -> 0S/1A/0
2. S -> 1A/0B , A -> 0A/1 , B -> 0A/1B/0

Exercise 4. Study equivalence between FA/NFA/RE and RLG and solve the following problems.

1. FA contains even no. of a's.
2. NFA contains strings ends with ba.
3. b*a(ba)*bb

Exercise 5. Study equivalence between FA/NFA/RE and LLG and solve the following problems.

1. (ba)*b(a+b)*aba*
2. b*a(a+b)*a(ab)*

Exercise 6. Study equivalence between RG (LLG/RLG) and RE and solve the following problems.

1. S -> 0A/11B , A -> 10A/00 , B -> 01B/1S
2. S -> 0AS/01B/01 , A -> 0A/11 , B -> 1B/S
3. S -> A01/B11 , a -> 01A/11 , B -> 10B/101
4. S -> 0S/1A , A -> 1B/01 , B ->1B/0

Exercise 7. Study equivalence between RLG and LLG and solve the following problems.

1. S -> 01B/0 , B -> 1B/11
2. S -> 01A/11B , A -> 0A/10 , B -> 0B/0

Exercise 8. Study equivalence between LLG and RLG and solve the following problems.

1. S -> A10/B01 , A-> A11/0 , B -> B101/1

Semester: 6 (Even-2026)
Subject: TOC
Assignment Topic: Normal Forms
Targeted CO: CO3
Date: 08/04/2026
Submission Deadline: 22/4/2026

Instructions:
Do as directed.

Prerequisites: Knowledge of PDA and CFG

Step 1. Study Types of Normal Form

- Chomsky Normal Form (CNF)
- Greibach Normal Form (GNF)

Step 2. Solve the exercises given below.

Exercise 1. Write down the definition of CNF and GNF with respect to CFG.

Exercise 2. Construct CNF for the following grammars.

3. S -> aAb/bA , A -> a/aB , B -> b/Ba
4. S -> aSb/ab
5. S -> aSa/bSb/c

Exercise 3. Construct GNF for the following grammars.

1. S -> aAb/Ba , A -> aB , B -> b/ε
2. S -> aSa/ab
3. S -> ABAC , A -> aA/ε , B -> bB/ε , C -> c